

check_pg_missing_vias

NAME

check_pg_missing_vias

Checks missing vias within the power and ground network.

SYNTAX

```
status check_pg_missing_vias
      [-nets pg_nets]
      [-within_bbox bboxes]
      [-via_rule_file rule_file_name]
      [-output_file output_file_name]
      [-write_default_via_rule_file]
      [-shape_use shape_use_list]
      [-ignore_shield_route]
      [-ignore_small_intersections]
      [-honor_routing_blockage]
      [-filter_by_drc]
```

Data Types

pg_nets	collection
rule_file_name	string
output_file_name	string
shape_use_list	list

ARGUMENTS

-nets pg_nets

Specifies the PG nets to be checked. By default, the command checks all PG nets.

-within_bbox bboxes

Specifies a bounding box or a list of bounding boxes within which you can check the missing PG vias.

-via_rule_file rule_file_name

Specifies the via rule file which includes all the via rules to be checked. By default, the command checks vias between adjacent orthogonal layers.

The via rule file can include multiple via rules.

The supported rules are

- o stack_via: Checks for stacked vias.
- By default (false), this rule is not checked.
- o parallel: Checks for vias between parallel layers.
- When you specify this rule, you must also specify the pitch with the pitch attribute.
- By default (false), this rule is not checked.
- o pitch Specifies the via pitch for parallel wires.
- o macro_pin: Checks for vias connecting to macro or power-switch PG pins.
- By default (false), this rule is not checked.
- o terminal: Checks for vias from terminal in upper layer to wires in lower layer
- By default (false), this rule is not checked.

The via rule format is as follows:

```
viaRuleName {
  bottom_layer: bottom_metal_layer_name (required)
  top_layer: top_metal_layer_name (required)
  stack_via: true | false
  parallel: true|false
  pitch: number
  macro_pin: true | false
  terminal: true | false
}
```

The following example checks vias between M2 and M7, including

stacked vias and macro pins.

```
viaRule1 {  
    bottom_layer: M2  
    top_layer:    M7  
    stack_via:    true  
    macro_pin:    true  
!}
```

-output_file output_file_name

Specifies the output file name. If specified, the tool writes detailed information about missing vias to the file.

-write_default_via_rule_file

Writes the default via rule to defaultViaRule.txt.

-shape_use

If specified, the command will only check vias against shapes with specified shape_use in the option. If not specified, the command will check vias against shapes with all shape_use.

-ignore_shield_route

If specified, the command will not check vias against shapes with shape use shield_route.

-ignore_small_intersections

If specified, the command will not report error if the via site is smaller than the minimum contact code on the layer.

-honor_routing_blockage

If specified, the command will not report error if the via site is covered by routing blockages.

-filter_by_drc

If specified, the command will not report error if the via site is found in the error browser as a DRC error by a command using the -show_phantom option.

DESCRIPTION

This command checks for missing vias between overlapping regions of different metal layers. The error info can be viewed from error browser or written to a file. You can define via rules to be checked by writing the rules to a file and importing the file with the -via_rule_file option.

EXAMPLES

The following example checks the vias for the VSS net based on rules defined in viaRule.txt.

```
prompt> check_pg_missing_vias -nets VSS -within_bbox {{900 900} {1100 1100}} -via_rule_file viaRule.txt  
Check net VSS vias...  
Number of missing vias: 0  
Checking net VSS vias took 0 seconds.
```

SEE ALSO

[create_pg_vias\(2\)](#)
[compile_pg\(2\)](#)

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